

Project Demo Planning

Date	15 March 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Slides	Presenting the core problem of flash floods, socio-economic impacts, system architecture, and tech stack overview.	2 mins	Team Lead / Presenter
2	Geospatial UI Map Walkthrough	Demonstrating the Leaflet/Mapbox base layer, overlay toggling, and interactive historical flood layer rendering.	2 mins	Frontend Engineer
3	AI Prediction Core Showcase	Showing real-time execution of the PyTorch UNet model with Sentinel-2 image inputs and displaying overlay prediction IoU metrics.	3 mins	Machine Learning Lead
4	Live Sensor Telemetry Run	Injecting simulated streamflow/precipitation data spikes into the live FastAPI broker to update dashboard widgets on the fly.	2 mins	Backend & IoT Dev
5	Twilio Automated Alerting	Triggering a critical flood warning live demo and receiving immediate automated geo-fenced SMS alerts on real mobile devices.	2 mins	Backend Developer
6	Summary & Open Q&A	Closing remarks, summarizing development accomplishments, discussing future scalability, and answering jury panel questions.	4 mins	All Team Members

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Highlight how modern satellite imagery combined with deep learning fills critical gaps left by conventional radar systems.
2	Solution Overview	Briefly showcase our real-time processing pipeline (Sentinel-2 Data → UNet Model → FastAPI Server → PostgreSQL/PostGIS).
3	Live Feature Demonstration	Strictly stick to the live test telemetry trigger; have backup recorded video frames ready just in case of internet latency.
4	Q&A Session	Prepare concise architectural justifications regarding model precision, inference times, and cellular dispatch scaling limits.